Nedim Omanović

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Profile

Computer Science and Informatics graduate (BSc) from the Faculty of Electrical Engineering, University of Sarajevo, focused on backend and full-stack development. Hands-on experience building and deploying web applications using TypeScript, Node.js, Express, Next.js, and SQL. Experienced with REST APIs, authentication, database integration, and agile team development. Interested in backend architecture, scalable systems, and cloud technologies.

Education

Faculty of Electrical Engineering, University of Sarajevo
Bachelor of Science (BSc), Computer Science and Informatics 2023 – 2026
Bachelor's Thesis: **Layout Algebra** — constraint-based DSL layout editor for representing web layouts through semantic relationships, mathematical constraints, variables, and operators.

Experience

Student Intern May 4 – May 29, 2026
Central Bank of Bosnia and Herzegovina, Sarajevo

- Participated in creating phishing attack scenarios and simulations and preparing documentation for security incident response procedures (Incident Response Plans)
- Analyzed suspicious network traffic and packets using Wireshark as part of network-security analysis
- Participated in verification of Oracle RMAN backup procedures and monitoring of Oracle Data Guard replication for backup, recovery, and high-availability processes
- Used Oracle Enterprise Manager (OEM) to monitor database performance, resource utilization, and security alerts
- Gained practical exposure to enterprise IT infrastructure, information security, databases, networking, and technical support in a regulated financial institution

Technical Skills

Programming Languages	TypeScript, JavaScript, Kotlin, C#, C, C++
Backend & APIs	Node.js, Express, REST APIs, Prisma ORM, Socket.IO, Zod
Frontend	React, Next.js, HTML, CSS, Tailwind CSS, Vite
Databases & BaaS	SQL, MySQL, Supabase, PostgreSQL
Cloud & Deployment	Cloudflare Pages, Vercel, Docker, Docker Compose,
Mobile	Android, Kotlin, Jetpack Compose
Tools	Git, GitHub, Postman, Wireshark, VS Code, Visual Studio, Android Studio, Jira, LaTe, Azure DevOps

Languages

Bosnian — Native English — C1

Key Projects

Layout Algebra — DSL Layout Editor *(Bachelor's Thesis)* GitHub Live
React, TypeScript, Vite, kiwi.js

- Designed and implemented a specialized domain-specific language (DSL) for describing web layouts using semantic relationships, variables, operators, elements, and constraints

- Built the complete language-processing pipeline from scratch: tokenization, parsing, AST construction, and interpretation of DSL commands
- Integrated the *kiwi.js* Cassowary solver for real-time resolution of mathematical layout constraints and automatic element positioning
- Implemented bidirectional synchronization between the DSL editor and visual canvas, plus CRUD, hierarchical elements, undo/redo history, and resizable panels
- Built an export system translating constraints into production-ready CSS using Absolute positioning, Flexbox, and CSS Grid

Service Intervention Management System *(Team Project, 8 members)* GitHub Live
Next.js, React, TypeScript, Node.js, Express, Prisma ORM, MySQL, Socket.IO, Docker Compose

- Developed a full-stack service intervention platform in an 8-member team across 10+ agile sprints
- Built intervention creation/editing, audit logging, Zod validation, 24+ unit tests, file attachments, user blocking, SLA/priority management, overdue detection, and management analytics
- Implemented Keycloak authentication and role-based access control; contributed to UI/UX, sprint planning, Jira task distribution, documentation, and GitFlow

Bridal Room — E-commerce Website *(Freelance)* GitHub Live
Next.js 15, TypeScript, Tailwind CSS, Supabase, Cloudflare Pages

- Designed, developed, and deployed a full-stack e-commerce website for a bridal dress shop for a paying client
- Built product filtering, availability and category management, and a custom admin panel with authentication
- Integrated Supabase for PostgreSQL data and image storage and Resend API for email delivery

EMA Chiptuning — Business Website *(Freelance)* GitHub Live
HTML5, CSS3, Vanilla JavaScript

- Developed and deployed a production website for a paying chiptuning-service client
- Built a calculator covering 130+ vehicle families and 800+ engine configurations with service recommendations
- Implemented SEO features, WhatsApp integration, and Google Maps integration

Other Projects

- **Ribolov BiH** *(Next.js, TypeScript, Tailwind CSS)* — Fishing assistant with weather data, location-based recommendations, interactive maps, solunar calculations, species-specific scores, and fishing regulations. GitHub Live
- **Android News App** *(Kotlin, Jetpack Compose)* — News app using MVVM, Retrofit, Coroutines, REST API integration, category filtering, and Compose state management. GitHub
- **Faculty Management System** *(C#)* — Desktop CRUD application for student/faculty records with database integration. GitHub